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Week 3

# SMOKING-ASSOCIATED GENE EXPRESSION CHANGES IN LUNG ADENOCARCINOMA: ANALYSIS OF GSE10072

## I INTRODUCTION

Lung cancer remains the leading cause of cancer-related mortality worldwide, with incidence and death rates that continue to rise each year. Among its histological subtypes, lung adenocarcinoma is the most frequently diagnosed, occurring in both current and former smokers. Tobacco smoking is estimated to account for more than 90% of lung cancer cases, underscoring the central role of prolonged exposure to carcinogenic compounds in cigarette smoke in driving pulmonary carcinogenesis (Hecht, 2012). Such exposure promotes the accumulation of genetic mutations, induces oxidative stress, and disrupts normal gene regulation, thereby contributing to the transformation of healthy epithelial cells into malignant cells (Govindan *et al.*, 2012).

Although the epidemiological association between smoking and lung cancer has long been established, the precise molecular mechanisms linking tobacco exposure to transcriptional alterations that promote adenocarcinoma progression remain incompletely understood. Transcriptomic approaches provide a powerful strategy to investigate these mechanisms by identifying differentially expressed genes (DEGs) between tumor and non-tumor lung tissues, as well as between smokers and non-smokers. Microarray technology, particularly the Affymetrix Human Genome U133A platform, has been widely utilized to generate comprehensive gene expression profiles across multiple cancer types (Irizarry *et al.*, 2003).

The publicly available dataset GSE10072, deposited in the Gene Expression Omnibus (GEO), contains gene expression profiles from 135 fresh-frozen lung adenocarcinoma samples and paired non-involved lung tissues, accompanied by biochemically validated smoking status information. The original study identified a smoking-associated gene expression signature enriched for genes involved in cell cycle regulation and mitotic spindle formation, including NEK2, TTK, and PRC1. These genes significantly distinguished tumors from smokers and non-smokers ($p < 0.001$; fold change > 1.5). Moreover, elevated expression of NEK2 and TTK in non-tumor lung tissue was associated with an approximately threefold increased risk of mortality among smokers with lung adenocarcinoma. These findings highlight the importance of mitotic regulation and cell cycle–related pathways in smoking-induced lung carcinogenesis.

Based on this background, a bioinformatic analysis of the GSE10072 dataset was conducted to identify significantly upregulated and downregulated genes. The results were visualized using a volcano plot and a heatmap of the top 50 differentially expressed genes. Furthermore, functional enrichment analyses were performed using

Gene Ontology (GO) and KEGG pathway databases to determine the biological processes and signaling pathways most strongly associated with smoking-related lung adenocarcinoma. This integrative approach is expected to provide deeper insight into the molecular mechanisms underlying smoking-driven lung carcinogenesis and to identify potential biomarkers and therapeutic targets..

## II METHODS

### 2.1 Study Design and Dataset Selection

Participants were allowed to select one case study or published dataset to analyze following the workflow introduced in the course module. The analysis could be conducted using either (1) the dataset and scripts provided in the module or (2) a publicly available dataset obtained independently from repositories such as the Gene Expression Omnibus (GEO), ArrayExpress, or other relevant databases.

In this study, the publicly available microarray dataset **GSE10072** was selected from the Gene Expression Omnibus (GEO). This dataset contains gene expression profiles derived from lung adenocarcinoma tissues and paired non-involved lung tissues, including validated information on smoking status. The dataset was chosen due to its relevance to smoking-associated lung carcinogenesis and its suitability for differential gene expression analysis.

### 2.2 Data Retrieval and Preprocessing

The dataset was downloaded directly from GEO using R and appropriate Bioconductor packages. After data importation, sample metadata were examined to confirm group classifications (e.g., tumor vs. non-tumor or smoker vs. non-smoker).

Quality control procedures were conducted to assess data consistency and detect potential outliers. The distribution of expression values across samples was visually inspected using boxplots and summary statistics. Because the dataset was generated using Affymetrix microarray technology, normalization procedures were applied to reduce technical variation and ensure comparability between samples.

Probe identifiers were mapped to gene symbols where necessary to facilitate downstream biological interpretation.

### 2.3 Differential Gene Expression Analysis

Differential gene expression analysis was performed to identify genes significantly upregulated or downregulated between predefined comparison groups. A linear modeling approach was applied, followed by empirical Bayes moderation to improve variance estimation.

To control for false-positive findings due to multiple testing, p-values were adjusted using the False Discovery Rate (FDR) method. Genes were considered significantly differentially expressed if they met the predefined thresholds of adjusted p-value $< 0.05$ and a meaningful fold-change cutoff.

The results were visualized using a volcano plot, displaying log fold change against –log10 adjusted p-value to clearly distinguish significantly upregulated and downregulated genes.

### 2.4 Visualization of Top Differentially Expressed Genes

The top 50 differentially expressed genes (DEGs) were selected based on statistical significance and magnitude of fold change. These genes were visualized using a heatmap to illustrate expression patterns across samples and to highlight clustering patterns between experimental groups.

Hierarchical clustering was applied to both genes and samples to evaluate similarity patterns and potential biological grouping.

### 2.5 Functional Enrichment Analysis

To investigate the biological relevance of the identified DEGs, functional enrichment analyses were conducted.

Gene Ontology (GO) enrichment analysis was performed to identify significantly overrepresented biological processes, molecular functions, and cellular components. Additionally, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis was conducted to determine signaling pathways significantly associated with the differentially expressed genes.

Enrichment results were visualized using appropriate graphical representations such as bar plots or dot plots to facilitate interpretation.

### 2.6 Analytical Framework

All analytical procedures followed the standardized workflow introduced in the course module, including dataset selection, preprocessing, statistical testing, visualization, and biological interpretation. This ensured methodological consistency and reproducibility while allowing flexibility in dataset selection.

## III RESULT AND DISCUSSION

The overall transcriptomic profiling of the GSE10072 dataset demonstrates a consistent and biologically meaningful pattern across multiple levels of analysis. Initial evaluation of global expression distributions shows that tumor and normal samples share highly comparable density curves, as illustrated in Figure 1a, indicating that the normalization process effectively reduced technical variation. Similarly, inspection of per-sample variability through boxplot visualization in Figure 1b reveals aligned medians and relatively uniform interquartile ranges across all samples. Such consistency is essential in microarray analysis, as proper normalization ensures that observed differences arise from biological variation rather than systematic bias (Bolstad *et al.*, 2003). Quality assessment approaches that include boxplot and

distributional inspection are widely recommended prior to differential expression testing (Kauffmann *et al.*, 2009).

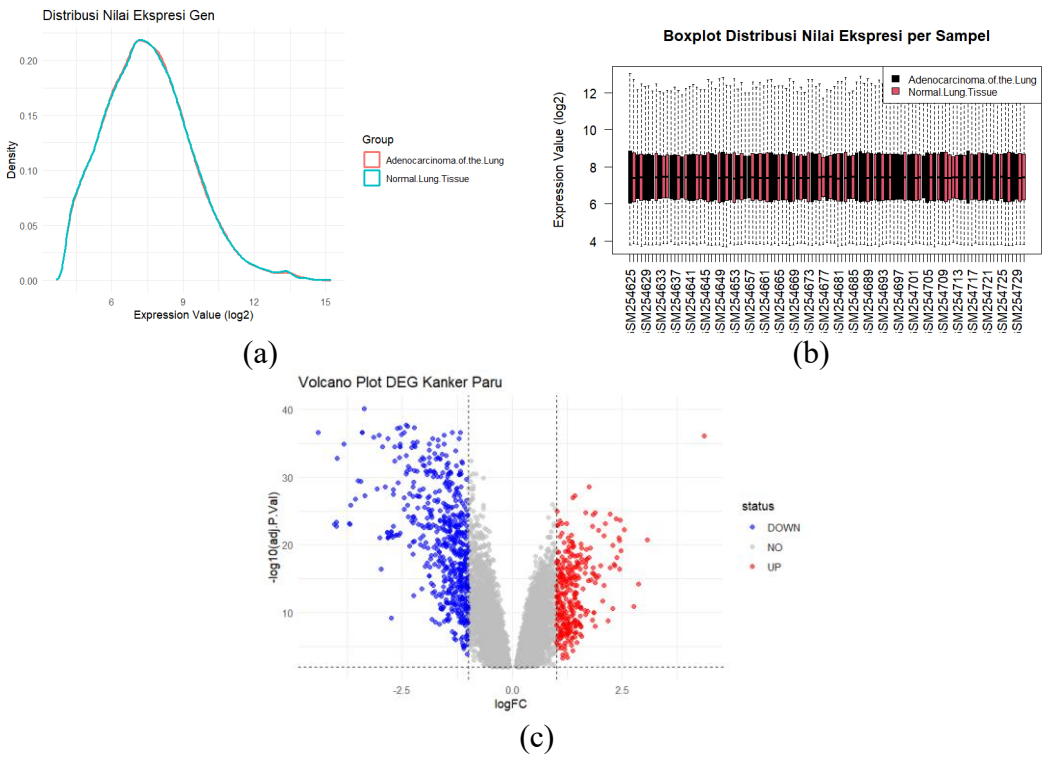



**Figure 1** (a) Density distribution of normalized gene expression values across samples in the GSE10072 dataset, (b) Boxplot of normalized gene expression levels in the GSE10072 dataset, (c) Volcano plot of differentially expressed genes (DEGs) in the GSE10072 dataset

Once data quality is established, gene-level comparisons reveal substantial transcriptional differences between adenocarcinoma and normal lung tissue. This is clearly visualized in the volcano plot shown in Figure 1c, where a subset of genes exceeds both fold-change and statistical significance thresholds. The characteristic volcano distribution reflects the biological principle that only specific gene subsets, rather than the entire transcriptome, are dysregulated in cancer. Visualization techniques that integrate magnitude and statistical confidence, such as volcano plots, are standard in differential expression studies because they allow rapid identification of biologically meaningful candidates (Cui & Churchill, 2003). The presence of both upregulated and downregulated genes supports the concept that tumorigenesis involves coordinated activation of proliferative pathways alongside repression of genes associated with normal tissue homeostasis.

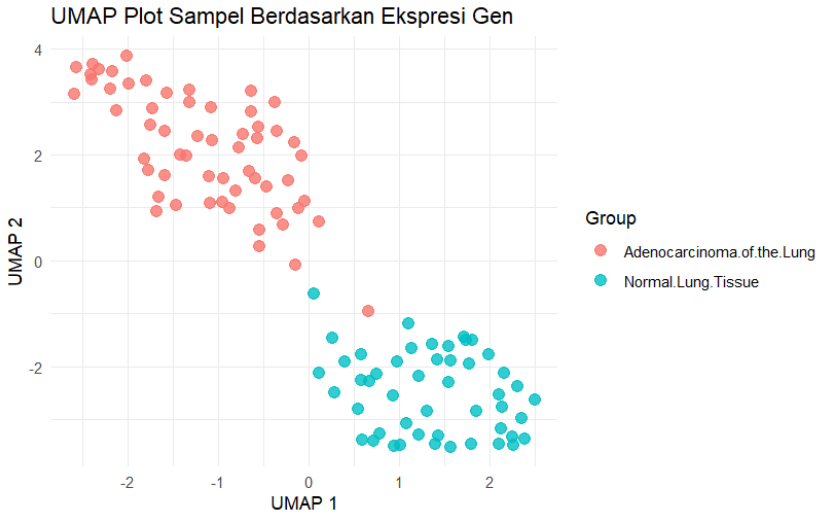



**Figure 2** UMAP plot of differentially expressed genes (DEGs) in the GSE10072 dataset

Beyond individual gene changes, the global structure of transcriptomic variation further reinforces disease-specific separation. Dimensionality reduction analysis demonstrates that samples cluster distinctly according to biological condition, as seen in the UMAP projection in Figure 2. The clear segregation between adenocarcinoma and normal lung tissues indicates that transcriptomic reprogramming in cancer is sufficiently robust to distinguish disease states at the whole-transcriptome level. UMAP has become a powerful method for visualizing complex biological datasets because it preserves both local and global relationships in high-dimensional data (McInnes *et al.*, 2018). The distinct clustering pattern observed here is consistent with large-scale molecular profiling efforts showing that lung adenocarcinoma possesses characteristic gene expression signatures differentiating it from non-tumor lung tissue (The Cancer Genome Atlas Research Network, 2014).

Taken together, the consistent distribution patterns in Figures 1 and 2 confirm high data quality, the differential expression pattern visualized in Figure 3 identifies targeted gene dysregulation, and the clustering structure observed in Figure 4 demonstrates global transcriptomic separation between disease states. This multi-level coherence strengthens confidence that the identified molecular differences reflect true biological processes underlying lung adenocarcinoma rather than technical artifacts.

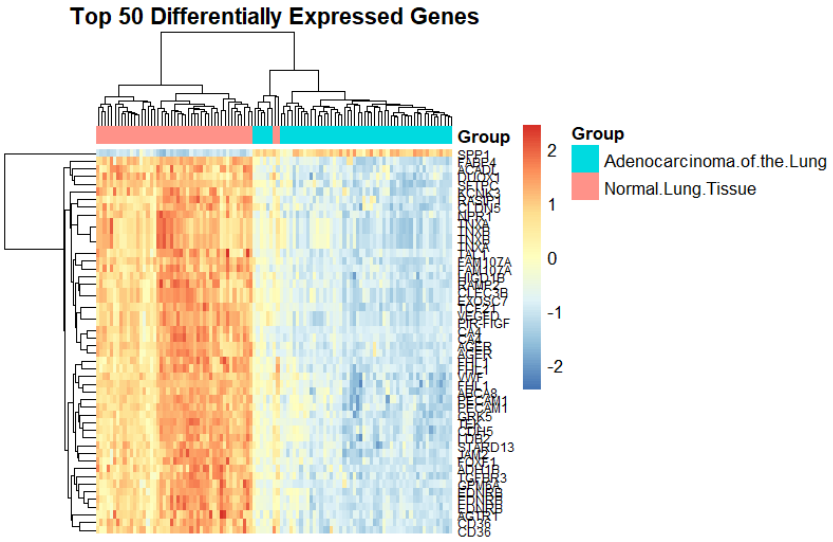



**Figure 3** Heatmap of differentially expressed genes (DEGs) in the GSE10072 dataset

The heatmap of the top 50 differentially expressed genes (DEGs) demonstrates a clear and well-defined segregation between lung adenocarcinoma tissues and normal lung tissues. Hierarchical clustering reveals two major clusters corresponding to tumor and normal samples, indicating consistent and coordinated transcriptional differences across the dataset. Tumor samples predominantly display elevated expression levels of several genes (represented in red), while a substantial subset of genes is downregulated (represented in blue) compared to normal lung tissue.

This distinct clustering pattern suggests that lung adenocarcinoma possesses a characteristic transcriptomic signature that differentiates malignant from non-malignant tissue. Such transcriptional divergence has been consistently reported in lung cancer molecular profiling studies, where tumor tissues exhibit substantial alterations in gene regulatory networks compared to normal lung epithelium (Rodenhuis & Slebos, 2010). The observed gene expression changes are unlikely to represent random variation; rather, they reflect coordinated dysregulation of biological pathways central to tumor development.

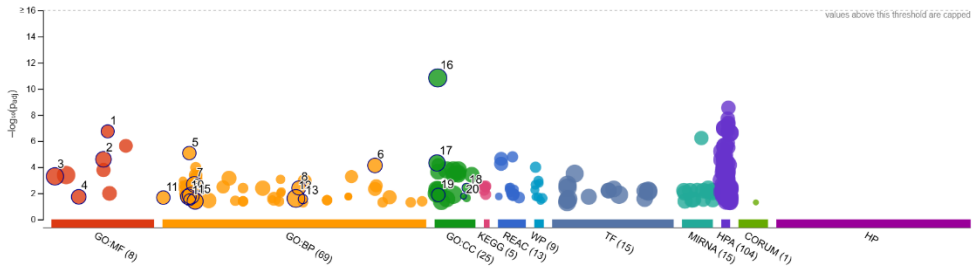
Moreover, integrative genomic analyses have demonstrated that transcriptomic reprogramming in lung adenocarcinoma involves widespread modulation of genes related to proliferation, survival signaling, and cellular microenvironment interactions (Zhu *et al.*, 2017). The upregulated genes observed in this heatmap likely contribute to enhanced proliferative capacity and tumor progression, whereas downregulated genes may indicate the loss of normal lung epithelial differentiation and functional specialization. This coordinated pattern aligns with the concept that cancer progression is driven by systematic alterations in gene networks rather than isolated gene events.

Importantly, clustering-based visualization such as heatmaps provides biologically meaningful insight into transcriptomic organization, enabling the identification of gene modules that behave similarly across samples (Hastie *et al.*, 2009). The robust separation observed in this analysis supports the reliability of the

identified DEGs and highlights their potential utility in downstream applications, including prognostic modeling and therapeutic target discovery.

Furthermore, previous integrative studies have shown that gene expression signatures derived from lung adenocarcinoma datasets can correlate with clinical outcomes and may serve as biomarkers for patient stratification (Bossé *et al.*, 2019). Therefore, the transcriptional patterns identified in this study not only validate the presence of molecular heterogeneity between tumor and normal tissues but also provide a strong foundation for subsequent prognostic and translational analyses.

Overall, the heatmap confirms the presence of a distinct and coordinated transcriptional landscape in lung adenocarcinoma, reinforcing the biological relevance of the identified gene signature and supporting its suitability for advanced bioinformatic investigation. Differential expression analysis of the GSE10072 dataset identified significantly up- and down-regulated genes between lung adenocarcinoma and normal lung tissues. To understand the biological meaning of these changes, GO and KEGG enrichment analyses were subsequently performed to interpret the results at the functional and pathway levels.




| ID | Source | Term ID | Term Name | $p_{adj}$ (query_1) |
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| 16 | GO:CC | GO:0005737 | cytoplasm | $1.464\times10^{-11}$ |
| 1 | GO:MF | GO:0045296 | cadherin binding | $1.823\times10^{-7}$ |
| 5 | GO:BP | GO:0007059 | chromosome segregation | $8.427\times10^{-6}$ |
| 2 | GO:MF | GO:0042802 | identical protein binding | $2.513\times10^{-5}$ |
| 17 | GO:CC | GO:0005615 | extracellular space | $4.781\times10^{-5}$ |
| 6 | GO:BP | GO:1901135 | carbohydrate derivative metabolic process | $7.412\times10^{-5}$ |
| 3 | GO:MF | GO:0003824 | catalytic activity | $5.079\times10^{-4}$ |
| 7 | GO:BP | GO:0008283 | cell population proliferation | $2.100\times10^{-3}$ |
| 18 | GO:CC | GO:0098644 | complex of collagen trimers | $3.852\times10^{-3}$ |
| 8 | GO:BP | GO:0051301 | cell division | $3.938\times10^{-3}$ |
| 19 | GO:CC | GO:0005819 | spindle | $1.348\times10^{-2}$ |
| 9 | GO:BP | GO:0007275 | multicellular organism development | $1.546\times10^{-2}$ |
| 20 | GO:CC | GO:0097125 | cyclin B1-CDK1 complex | $1.673\times10^{-2}$ |
| 4 | GO:MF | GO:0016301 | kinase activity | $1.878\times10^{-2}$ |
| 10 | GO:BP | GO:0006915 | apoptotic process | $1.884\times10^{-2}$ |
| 11 | GO:BP | GO:0000280 | nuclear division | $2.140\times10^{-2}$ |
| 12 | GO:BP | GO:0050896 | response to stimulus | $2.497\times10^{-2}$ |
| 13 | GO:BP | GO:0051988 | regulation of attachment of spindle microtubules t... | $2.755\times10^{-2}$ |
| 14 | GO:BP | GO:0007088 | regulation of mitotic nuclear division | $3.167\times10^{-2}$ |
| 15 | GO:BP | GO:0009056 | catabolic process | $4.058\times10^{-2}$ |

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**Figure 4** Gene Ontology (GO) enrichment analysis of up-regulated genes in the GSE10072 dataset

The up-regulated gene set is strongly enriched in cell cycle–related processes, including *chromosome segregation*, *cell division*, *nuclear division*, and structural components such as *spindle* and *cyclin B1–CDK1 complex*. Enrichment in *cell population proliferation* and *kinase activity* further supports enhanced mitotic activity.

These findings are consistent with the fundamental hallmark of cancer: sustained proliferative signaling. The Cyclin B1–CDK1 complex is a key regulator of the G2/M transition, and its overactivation has been widely reported in multiple cancers, including lung cancer (Malumbres & Barbacid, 2009). Additionally, enrichment in *cadherin binding* and extracellular components suggests altered cell–cell adhesion dynamics, which contribute to tumor progression and invasiveness. Disruption of adhesion regulation is a recognized mechanism in cancer development (Hanahan & Weinberg, 2011).

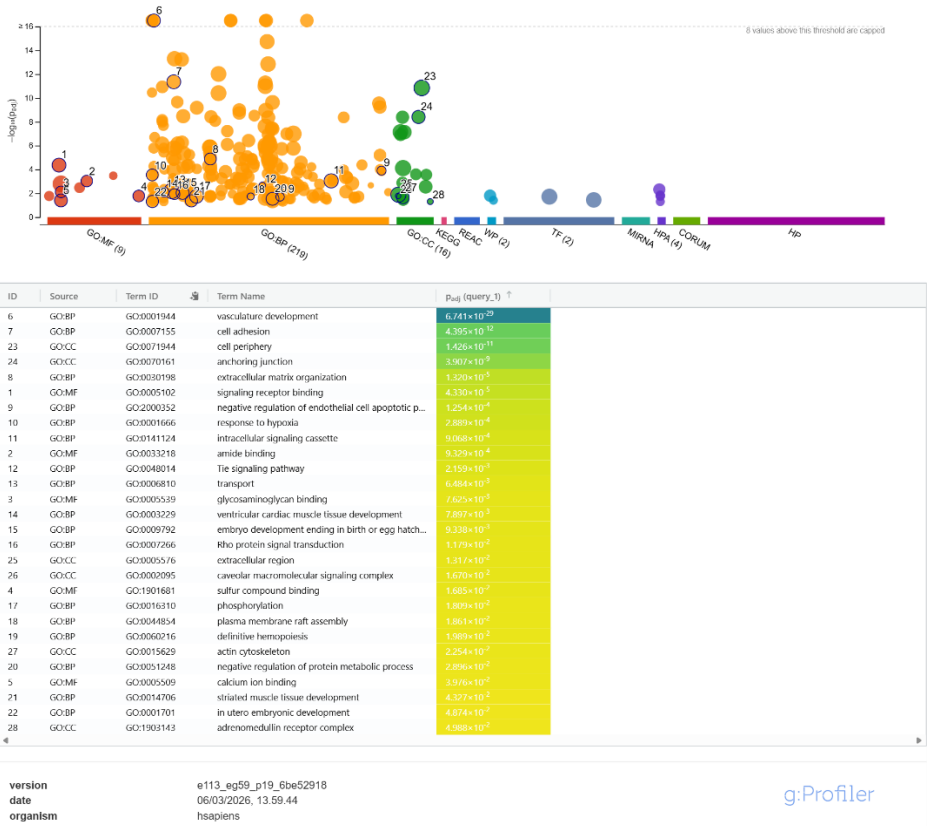



**Figure 5** Gene Ontology (GO) enrichment analysis of down-regulated genes in the GSE10072 dataset

In contrast, the down-regulated gene set shows significant enrichment in processes such as *vasculature development*, *cell adhesion*, *extracellular matrix organization*, *response to hypoxia*, and *Rho protein signal transduction*. Down-regulation of extracellular matrix (ECM) organization and adhesion pathways indicates disruption of normal tissue architecture. Under physiological conditions, ECM components maintain structural integrity and regulate signaling, whereas in cancer, their dysregulation contributes to tumor microenvironment remodeling (Lu *et al.*, 2012). Reduced enrichment in hypoxia response–related processes may reflect altered oxygen-sensing mechanisms within tumor tissues. Hypoxia signaling, particularly

mediated by HIF pathways, plays a crucial role in solid tumor progression, including lung adenocarcinoma (Semenza, 2012). Thus, GO down-regulation highlights impaired tissue homeostasis and microenvironmental alterations accompanying malignant transformation.

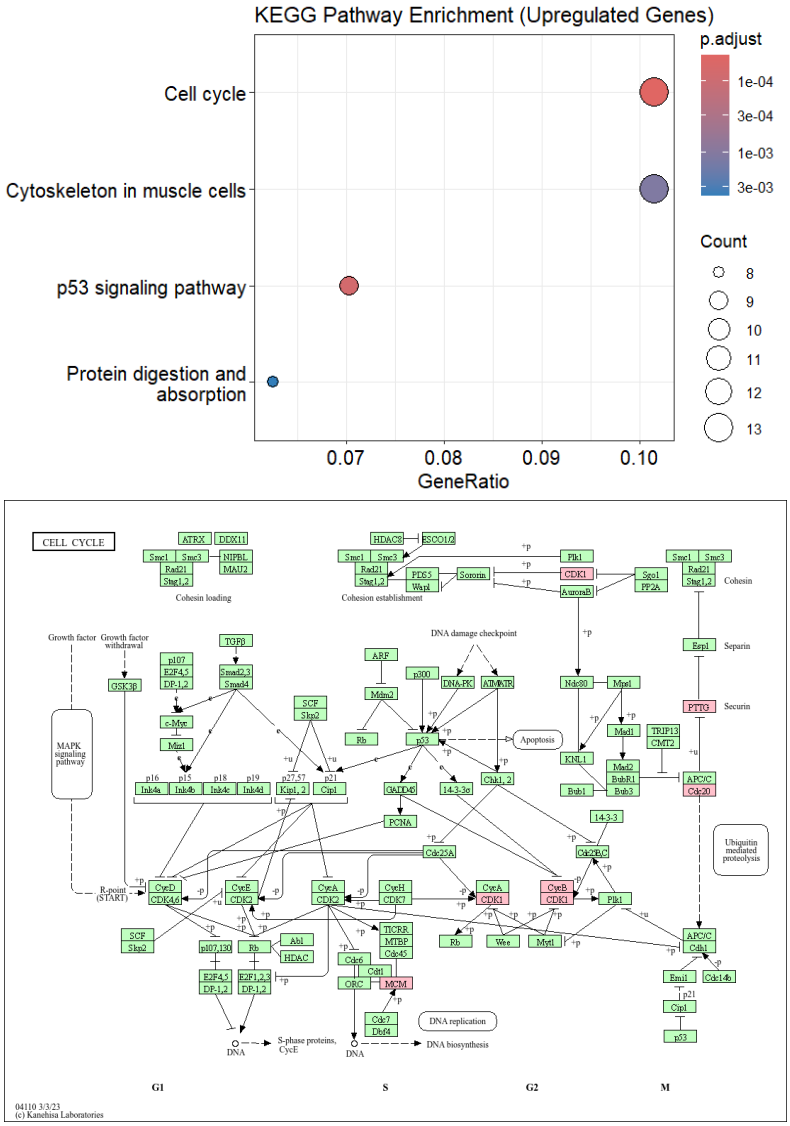
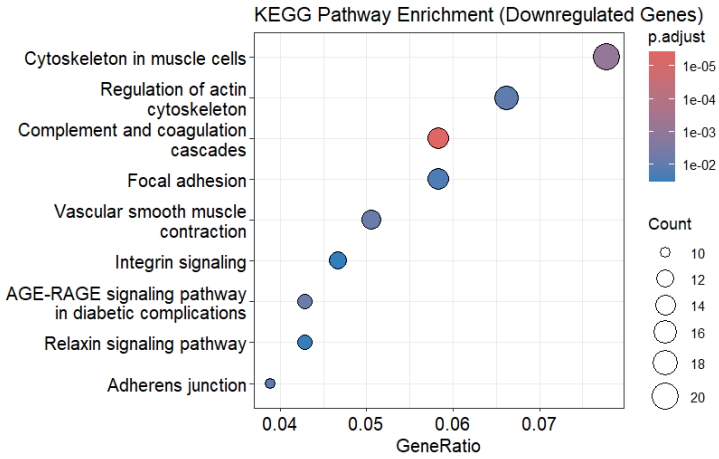



**Figure 6** KEGG pathway enrichment analysis of up-regulated genes in the GSE10072 dataset

KEGG pathway enrichment analysis of the upregulated genes revealed several pathways associated with increased cellular proliferation and tumor progression. Among the enriched pathways, the cell cycle pathway showed the highest enrichment,

suggesting that many upregulated genes are involved in regulating cell division and mitotic progression. Dysregulation of the cell cycle is a well-known hallmark of cancer and plays a critical role in uncontrolled tumor cell proliferation (Hanahan & Weinberg, 2011). Previous studies have also shown that genes related to cell cycle regulation are frequently overexpressed in lung adenocarcinoma and contribute to tumor initiation and progression (Wang *et al.*, 2021). In addition, enrichment of the p53 signaling pathway indicates the involvement of genes associated with DNA damage response and cell cycle checkpoint regulation, which are essential mechanisms for maintaining genomic stability but are often disrupted in cancer cells (Levine, 2020).

Furthermore, pathways related to cytoskeleton organization were also enriched among the upregulated genes. Changes in cytoskeletal dynamics can influence cellular structure, migration, and invasion in cancer cells. These structural alterations support tumor progression and metastasis by enabling cancer cells to modify their morphology and interactions with the surrounding microenvironment (Fife *et al.*, 2014).



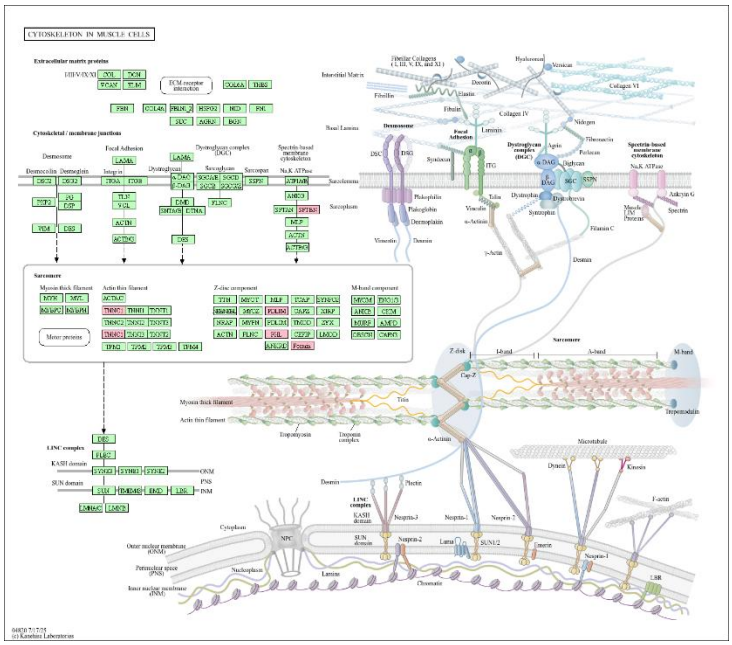



**Figure 7** KEGG pathway enrichment analysis of down-regulated genes in the GSE10072 dataset

In contrast, KEGG enrichment analysis of the downregulated genes highlighted pathways associated with cell adhesion, cytoskeletal organization, and vascular regulation, including focal adhesion, integrin signaling, and adherens junction pathways. These pathways play crucial roles in maintaining cell–cell and cell–extracellular matrix interactions in normal tissues. Downregulation of adhesion-related pathways may disrupt tissue architecture and facilitate tumor cell invasion and metastasis (Hamidi & Ivaska, 2018). Similar findings have been reported in lung adenocarcinoma studies where reduced expression of adhesion-related genes contributes to cancer progression and metastatic potential (Zhang *et al.*, 2025).
Overall, these enrichment results indicate that lung adenocarcinoma is characterized by increased activity in pathways related to cell proliferation and cell cycle regulation, while pathways involved in cell adhesion and tissue structural integrity tend to be suppressed. This molecular imbalance reflects the biological reprogramming that occurs during tumor development, where cancer cells gain proliferative advantages while losing mechanisms that maintain normal tissue organization.

## IV Conclusion

Transcriptomic analysis of the GSE10072 dataset shows clear differences in gene expression between lung adenocarcinoma tissues and normal lung tissues. Several genes were significantly upregulated and downregulated, and visualization using volcano plots, heatmaps, and UMAP demonstrated a clear separation between tumor and normal samples. Functional enrichment analysis indicates that upregulated genes are mainly involved in cell cycle regulation, cell division, and cellular proliferation, while downregulated genes are related to cell adhesion and extracellular matrix organization. These findings suggest that lung adenocarcinoma is associated with increased proliferative activity and disruption of normal tissue regulation at the transcriptomic level.

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